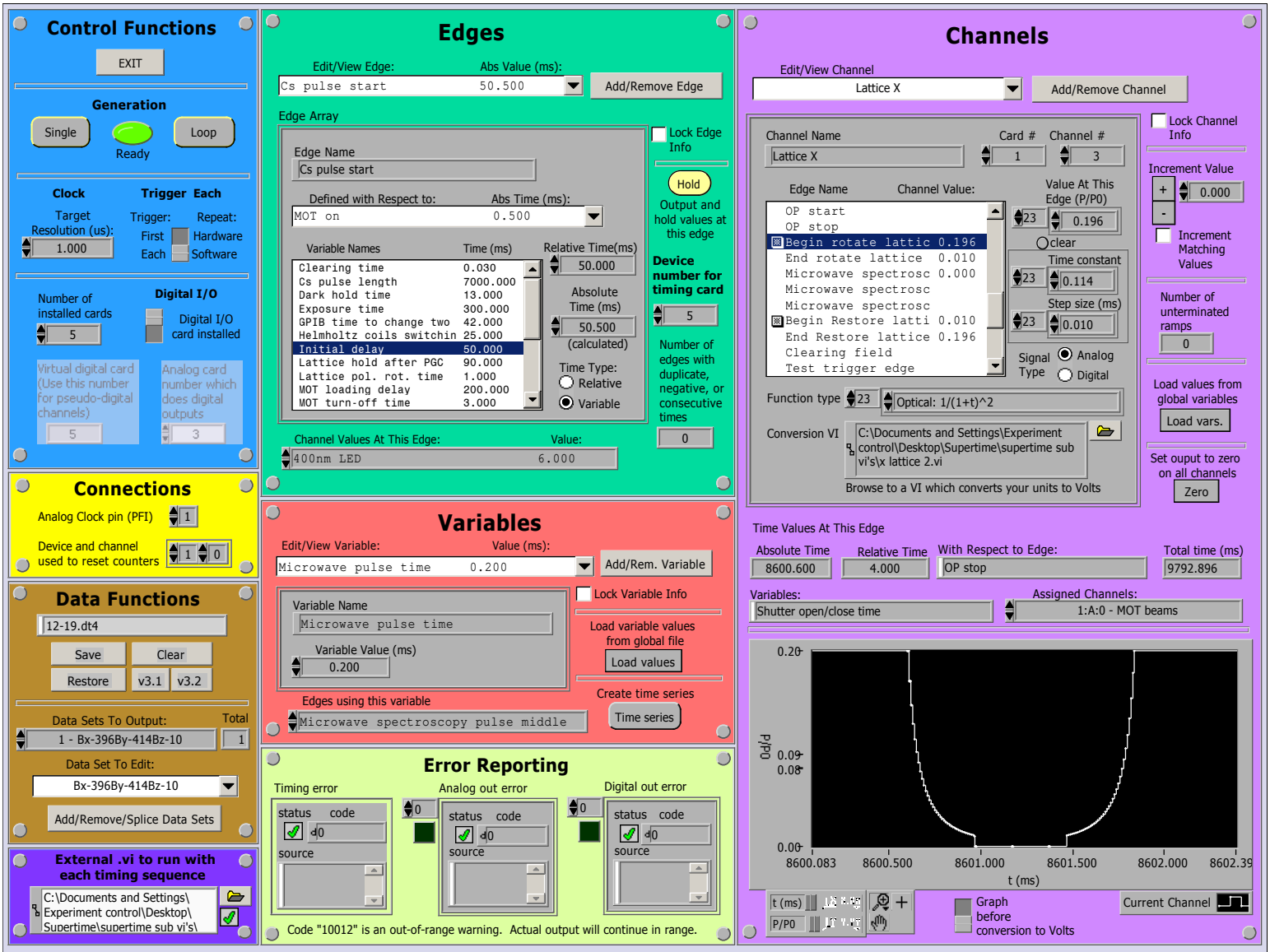

Control Functions
EXIT
Generation
Single
Loop
Ready
Clock
Target Resolution (us):
1.000
Trigger Each
Trigger:
First
Each
Repeat:
Hardware
Software
Number of installed cards
5
Digital I/O
Digital I/O card installed
Virtual digital card (Use this number for pseudo-digital channels)
5
Analog card number which does digital outputs
3
Connections
Analog Clock pin (PFI)
1
Device and channel used to reset counters
1
0
Data Functions
12-19.dt4
Save
Clear
Restore
v3.1
v3.2
Data Sets To Output:
1 - Bx-396By-414Bz-10
Total
1
Data Set To Edit:
Bx-396By-414Bz-10
Add/Remove/Splice Data Sets
External .vi to run with each timing sequence
C:\Documents and Settings\Experiment control\Desktop\Supertime\supertime sub vi's\
Edges
Edit/View Edge:
Cs pulse start
Abs Value (ms):
50.500
Add/Remove Edge
Edge Array
Edge Name
Cs pulse start
Defined with Respect to:
MOT on
Abs Time (ms):
0.500
Variable Names
Time (ms)
Clearing time 0.030
Cs pulse length 7000.000
Dark hold time 13.000
Exposure time 300.000
GPIB time to change two 42.000
Helmholtz coils switchin 25.000
Initial delay 50.000
Lattice hold after PGC 90.000
Lattice pol. rot. time 1.000
MOT loading delay 200.000
MOT turn-off time 3.000
Relative Time(ms)
50.000
Absolute Time (ms)
50.500
(calculated)
Time Type:
Relative
Variable
Lock Edge Info
Hold
Output and hold values at this edge
Device number for timing card
5
Number of edges with duplicate, negative, or consecutive times
0
Channel Values At This Edge:
400nm LED
Value:
6.000
Variables
Edit/View Variable:
Microwave pulse time
Value (ms):
0.200
Add/Rem. Variable
Variable Name
Microwave pulse time
Variable Value (ms)
0.200
Lock Variable Info
Load variable values from global file
Load values
Edges using this variable
Microwave spectroscopy pulse middle
Create time series
Time series
Error Reporting
Timing error
Analog out error
Digital out error
status
code
source
Code "10012" is an out-of-range warning. Actual output will continue in range.
Channels
Edit/View Channel
Lattice X
Add/Remove Channel
Channel Name
Lattice X
Card #
1
Channel #
3
Edge Name
Channel Value:
OP start
OP stop
Begin rotate lattic 0.196
End rotate lattice 0.010
Microwave spectrosc 0.000
Microwave spectrosc
Microwave spectrosc
Begin Restore latti 0.010
End Restore lattice 0.196
Clearing field
Test trigger edge
Value At This Edge (P/P0)
0.196
clear
Time constant
0.114
Step size (ms)
0.010
Signal Type
Analog
Digital
Function type
Optical: 1/(1+t)^2
Conversion VI
C:\Documents and Settings\Experiment control\Desktop\Supertime\supertime sub vi's\x lattice 2.vi
Browse to a VI which converts your units to Volts
Lock Channel Info
Increment Value
0.000
Increment Matching Values
Number of unterminated ramps
0
Load values from global variables
Load vars.
Set ouput to zero on all channels
Zero
Time Values At This Edge
Absolute Time
8600.600
Relative Time
4.000
With Respect to Edge:
OP stop
Total time (ms)
9792.896
Variables:
Shutter open/close time
Assigned Channels:
1:A:0 - MOT beams
P/P0
0.20
0.09
0.08
0.00
8600.083
8600.500
8601.000
8601.500
8602.000
8602.39
t (ms)
t (ms)
P/P0
Graph before conversion to Volts
Current Channel
Non-essential indicators
You may push the Esc key to exit the program if the Exit button is not visible
Set Name Array
Bx-396By-414Bz-10
times out array
actual clock freq.
1.00E+6
Sets To Run Array
analog volt
0.00
of scans
306028
points
9486849.
Buffer size Array
184
184
184
0
0
analog write data
channel
card
edge
channel
edge
type specifier VI Refnum (for type only)
ext. .vi error out
reference error
notifier error
status
code
source
analog taskIDs
200000
digital taskIDs
0
timing taskID
141000
digital write data
edge
card
edge
card
Channels on digital card are encoded as (up to) 32-bit integers for each card, with channel 0 the LSB and channel 31 the MSB.
data socket error